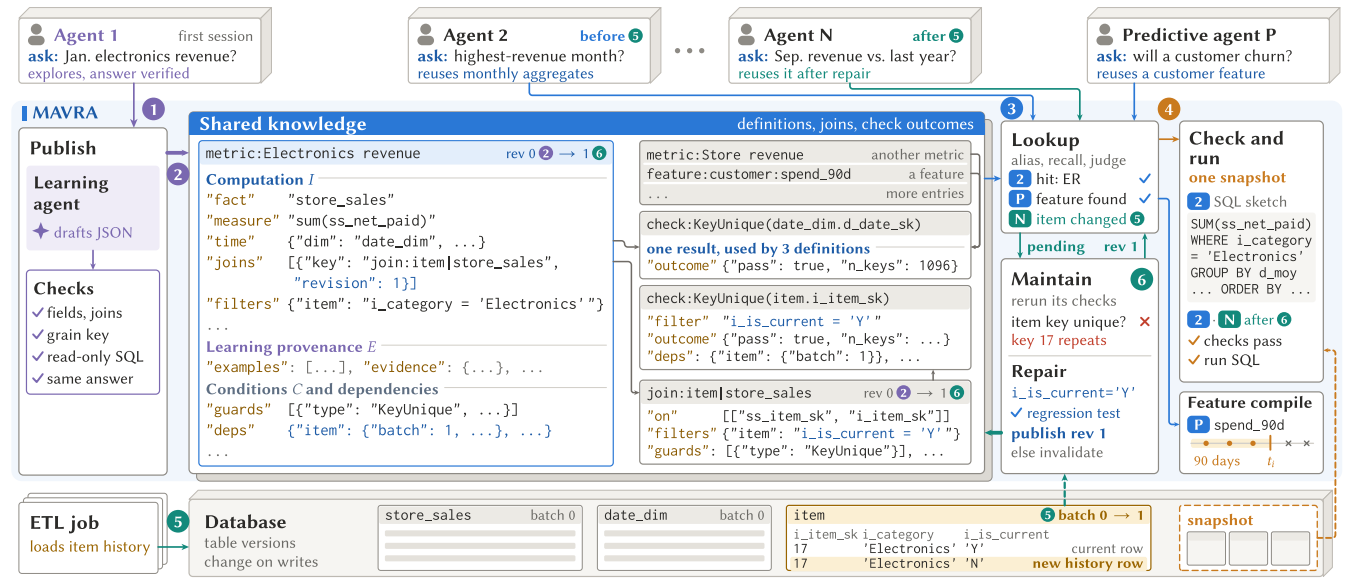

Agent 1
first session
ask: Jan. electronics revenue?
explores, answer verified
Agent 2
before 5
ask: highest-revenue month?
reuses monthly aggregates
Agent N
after 5
ask: Sep. revenue vs. last year?
reuses it after repair
Predictive agent P
ask: will a customer churn?
reuses a customer feature
MAVRA
1
Publish
Learning agent
drafts JSON
Checks
fields, joins
grain key
read-only SQL
same answer
Shared knowledge
definitions, joins, check outcomes
2
metric:Electronics revenue
rev 0 2 → 1 6
Computation I
"fact" "store_sales"
"measure" "sum(ss_net_paid)"
"time" {"dim": "date_dim", ...}
"joins" [{"key": "join:item|store_sales",
"revision": 1}]
"filters" {"item": "i_category = 'Electronics'"}
...
Learning provenance E
"examples": [...], "evidence": {...}, ...
Conditions C and dependencies
"guards" [{"type": "KeyUnique", ...}]
"deps" {"item": {"batch": 1, ...}, ...}
...
metric:Store revenue
another metric
feature:customer:spend_90d
a feature
...
more entries
check:KeyUnique(date_dim.d_date_sk)
one result, used by 3 definitions
"outcome" {"pass": true, "n_keys": 1096}
check:KeyUnique(item.i_item_sk)
"filter" "i_is_current = 'Y'"
"outcome" {"pass": true, "n_keys": ...}
"deps": {"item": {"batch": 1}}, ...
join:item|store_sales
rev 0 2 → 1 6
"on" [["ss_item_sk", "i_item_sk"]]
"filters" {"item": "i_is_current = 'Y'"}
"guards": [{"type": "KeyUnique"}], ...
3
Lookup
alias, recall, judge
2 hit: ER
P feature found
N item changed 5
pending rev 1
Maintain
6
rerun its checks
item key unique? ×
key 17 repeats
Repair
i_is_current='Y'
regression test
publish rev 1
else invalidate
4
Check and run
one snapshot
2 SQL sketch
SUM(ss_net_paid)
WHERE i_category
= 'Electronics'
GROUP BY d_moy
... ORDER BY ...
2 · N after 6
checks pass
run SQL
Feature compile
P spend_90d
90 days
t_i
ETL job
loads item history
5
Database
table versions
change on writes
store_sales
batch 0
date_dim
batch 0
item
5 batch 0 → 1
i_item_sk i_category i_is_current
17 'Electronics' 'Y' current row
17 'Electronics' 'N' new history row
snapshot